

MANTLE AI HACKATHON • TRACK: AI ALPHA & DATA

MANTLE SENTINEL

Behavioral Anomaly Detection for Smart Contracts

THE PROBLEM

\$3.4B stolen in crypto hacks in 2025



Signature-based tools miss novel attacks — they only know what they've seen



LLM monitors are slow, non-deterministic, and expensive at scale



New protocols have zero historical data to train on

THE SOLUTION

Every contract has a **behavioral fingerprint**.

Sentinel detects the moment it changes.

10,000

dimensions

0

training data

0

GPU required

HOW IT WORKS

T0–T5 Detection Pipeline

```
TX Stream → T0 Entropy Filter → T1 HDC Encoder → T2 Drift Signal → T3 Detector → T4 Attribution → T5 Z.ai → Alert
```

T1 5 features → 10,000-dim bipolar hypervector per window

T2 drift = max(Hamming distance, timing deviation)

T4 Feature ablation — algebraic attribution before any LLM call

⚡ **LLM is NEVER in the detection loop. Detection is deterministic, byte-for-byte.**

Z.AI INTEGRATION

Structured finding → **Z.ai GLM** → plain English brief

```
// Tier-4 structured finding: { "drift": 0.87, "attribution": {"selector": 0.62, "timing": 0.25},  
"episode_id": "ep-7a3f" } ↓ Z.ai GLM (glm-4.5-flash) "Unusual selector distribution shift detected. The  
contract received 62% more calls to an uncommon function selector in the last 50-tx window."
```



Strict template — may only restate Tier-5 structured findings



No API key → deterministic fallback (CI-safe)

LIVE DEMO RESULTS

Contract	Protocol	Health	Status
0x09bc4e...	USDC.e	83 / 100	✓ Healthy
0xcfA5aE...	Lendle Pool	81 / 100	✓ Healthy
0x78c1b0...	WMNT	74 / 100	⚠ Moderate
0x201eba...	USDT	70 / 100	⚠ Moderate
0xcda86a...	mETH	68 / 100	⚠ Moderate

4.3×
separation ratio

0
false positives

129
tests passing

ON-CHAIN PROOF

Immutable. Verifiable. **On Mantle Mainnet.**

```
SentinelAlertRegistry v2 0x0899E1507CFfeFF8620455721F5bd528Bb072187 Chain: Mantle Mainnet (5000) Alerts:  
getAlertCount() → on-chain Drift: driftScore stored ×10,000 Events: alertIndex for timeline  
reconstruction
```

Live TX: 0x086cf0...fa91c — mainnet block 96,680,154

DEVELOPER EXPERIENCE

One command. Full behavioral profile.

```
$ python -m sentinel scan 0x09bc4e... --min-health 60 Health Score: 83 / 100  Drift (p99): 0.730  
Alerts: 4 Report saved: bench/reports/0x09bc4e...json
```

GitHub Actions

Weekly health scan + manual trigger

Pre-commit

Push-stage health gate

Telegram

@MantleSentinelBot alerts

ROADMAP

NOW

- Single-contract monitoring
- CLI + Python SDK
- Telegram alerts
- On-chain alert anchoring
- CI/CD health gate

NEXT

- Multi-contract dashboard
- Confidence scoring
- Alert severity tiers
- Webhook integrations

FUTURE

- EVM multi-chain
- SaaS dashboard
- Insurance oracle
- MCP integration

MANTLE SENTINEL

Detect Before They Hit



mntsentinel.xyz



mntsentinel.xyz/dashboard



github.com/alexbelij/mantle-sentinel



[@MantleSentinelBot](https://t.me/MantleSentinelBot)